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 Milan Suryavanshi • modified 3 mins ago • 136 views

How to use LangChain Agent with tools in a Python project?

 Advice

python

artificial-intelligence

langchain

large-language-model

I'm trying to build a simple project using LangChain where an agent can decide when to call a tool (e.g., a calculator or API).

I understand basic chains, but I'm confused about:

- When to use a Chain vs an Agent
- How to properly define and register tools
- How the agent decides which tool to call

Here's what I've tried:

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```
from langchain.agents import initialize_agent, Tool
from langchain.llms import OpenAI

def my_tool(input):
    return "Processed: " + input

tools = [
    Tool(name="MyTool", func=my_tool, description="Processes input")
]

llm = OpenAI()

agent = initialize_agent(
    tools,
    llm,
    agent="zero-shot-react-description",
    verbose=True
)

response = agent.run("Process this text")
print(response)
```

The agent runs, but it doesn't seem to use the tool correctly.

Questions:

- How does the agent decide to use a tool?
- What's the correct way to structure tools for real projects?
- When should I use a Chain instead of an Agent?

Any guidance or examples would help.



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Ajeet Verma Apr 24 at 2:09

When to use a **Chain** vs an **Agent**

Use a Chain when:

- the steps are known ahead of time (like $A \rightarrow B \rightarrow C$)
- you don't need dynamic branching/tool choice
- you want predictability, speed, and easier debugging

and use an Agent when:

- you don't know which steps/tools are needed upfront
- tool choice depends on the user question
- you need iterative reasoning (tool call $\rightarrow$ observe $\rightarrow$ next step)

A practical rule would be:

- If you can write it as a deterministic pipeline $\rightarrow$ **Chain**
- If the pipeline depends on runtime decisions (and may vary per question) $\rightarrow$ **Agent**



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